Year 12 Maths - Using graphs

Similar questions - Intersections and quadratic models

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 Find the coordinates of the points of intersection of the curve $2xy + x^2 - 5 = 0$ and the line $x + y = 3$. **[5]**

2 A farmer models the extra yield, Y tonnes per hectare, of a crop when f kg per hectare of fertiliser is applied, by $Y = 3f - 0.025f^2$ for $0 \leqslant f \leqslant 120$.

(a) Find the amount of fertiliser that gives the greatest extra yield, and the greatest extra yield. **[4]**

(b) Suggest a reason why the model has an upper limit of 120 for f. **[1]**

(c) A second model $Y = 3f - 0.025f^2 + 10$ is proposed. Give a reason why it is not suitable. **[1]**

(d) A third model is $Y = 3.6f - 0.03f^2$ for $0 \leqslant f \leqslant 120$. Explain why this is a more sensible way to predict a larger extra yield. **[2]**

Question 1

$y = 3 - x$. $2x(3-x) + x^2 - 5 = 0$, so $6x - 2x^2 + x^2 - 5 = 0$, i.e. $x^2 - 6x + 5 = 0$ and $(x-1)(x-5) = 0$. $x = 1,\ y = 2$ and $x = 5,\ y = -2$. Points $(1, 2)$ and $(5, -2)$

Question 2

(a) Maximum at the line of symmetry $f = \dfrac{3}{2 \times 0.025} = 60$ kg per hectare. $Y = 180 - 90 = 90$ tonnes per hectare

(b) For $f > 120$ the model gives $Y < 0$, a negative extra yield, which makes no sense

(c) With no fertiliser ($f = 0$) the model gives $Y = 10$, but there should be no *extra* yield from no fertiliser

(d) It still gives $Y = 0$ at $f = 0$ and at $f = 120$, so it fits the limits, but its maximum is $3.6(60) - 0.03(60)^2 = 108 > 90$, a larger extra yield at the same fertiliser level